

# How Far Does a Near-Perfect Speech Emotion Recogniser Generalise?

Replication and Stress-Testing of a Lightweight Hand-Crafted-Feature Ensemble

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## Abstract

Chowdhury et al. (*Scientific Reports*, 2025) report 97.57% accuracy on RAVDESS and 100% on TESS for a lightweight averaging ensemble of a 1D-CNN and a CNN\_Bi-LSTM trained on hand-crafted features (ZCR, RMSE, MFCC, Chroma STFT). We re-implement the pipeline from the paper’s specification and stress-test it with four evaluation protocols, classical baselines, a gender-bias check and a feature ablation. With augmentation applied only to training clips, the ensemble reaches 81.9%; applying augmentation *before* the train-test split, so that 99.1% of test samples have an augmented copy in training, reproduces the paper’s level (96.5%). For speakers never seen in training, accuracy falls to  $58.5 \pm 1.9\%$ , and across corpora it is near chance (19.7% RAVDESS→TESS, 14.1% TESS→RAVDESS). All models are significantly more accurate for female than male actors (ensemble 66.2% vs. 50.7%,  $p < 0.001$ ; actor-level  $p = 0.001$ ). The architecture is a sound lightweight model, but its headline accuracy depends on the evaluation protocol; speaker-independent and cross-corpus evaluation should be the standard for reporting SER results.

## 1 Introduction

Speech emotion recognition (SER) classifies a speaker’s emotional state from the voice alone and is used in call-centre analytics, health screening and conversational agents. Chowdhury et al. [1] recently showed that a *lightweight* ensemble of a 1D-CNN and a CNN\_Bi-LSTM, trained on four classical hand-crafted features (ZCR, RMSE, MFCC and Chroma STFT), reaches 97.57% accuracy on RAVDESS and 100% on TESS, outperforming heavier spectrogram-based models. Such numbers are attractive for low-resource deployment, but they were obtained by training and testing on random splits of the *same* dataset, the paper does not state whether augmentation was applied before or after splitting, and the authors themselves list cross-dataset validation as future work [1, Sec. 7].

**Research question.** *Does the near-perfect accuracy of this lightweight model reflect emotion recognition that transfers to new clips, new speakers and new recording conditions, and is it equally accurate for male and female*

*speakers?* We re-implement the full pipeline from the paper’s specification and test six hypotheses:

- H1 Replication:** under the paper’s protocol our implementation approaches the reported RAVDESS accuracy.
- H2 Split order:** augmenting *before* splitting inflates accuracy, because augmented copies of a test clip then appear in training.
- H3 Unseen speakers:** accuracy drops substantially when test actors are never seen in training.
- H4 Unseen dataset:** accuracy collapses when training on RAVDESS and testing on TESS (and vice versa).
- H5 Gender bias:** speaker-independent accuracy differs between male and female actors.
- H6 Features:** MFCC carries most of the information; Chroma adds a smaller but positive contribution, as the paper’s LIME analysis suggests.

**Contributions.** (i) An open, reproducible re-implementation of the paper’s features, augmentation, architectures and training schedule; (ii) a controlled comparison of four evaluation protocols of increasing strictness, with classical baselines; (iii) the gender-bias check required by the course; (iv) a feature and augmentation ablation; (v) an error analysis explaining *why* performance changes between protocols.

## 2 Data

We use the two Kaggle datasets used by the paper [2, 3]. **RAVDESS** [4]: 1,440 speech clips (16-bit, 48 kHz) from 24 professional actors (12 male, 12 female, odd IDs male) speaking two fixed sentences in eight emotions at two intensities. **TESS** [5]: 2,800 clips (24.4 kHz) of the carrier phrase “Say the word \_\_\_\_” with 200 target words, spoken by two female actors aged 26 (YAF) and 64 (OAF) in seven emotions. Some distributed copies of both datasets contain the same files in duplicated folders, so our loader de-duplicates by filename to rule out train-test overlap.

Following the paper’s Table 1 (288 neutral RAVDESS clips), RAVDESS *calm* is merged into *neutral*, and TESS *pleasant surprise* is mapped to *surprised*, giving one 7-class label set for both corpora: neutral, happy, sad, angry, fearful, disgust, surprised. RAVDESS is balanced except neutral (288 vs. 192 per class); TESS is perfectly balanced (400 per class).

## 3 Methodology

### 3.1 Features and augmentation

All settings follow the paper’s Section 4 (Fig. 1). Audio is resampled to 22,050 Hz and 2.5 s are loaded from an offset of 0.6 s. With a 2,048-sample frame and 512-sample hop this gives 108 frames. Per frame we compute ZCR (1), RMSE (1), 20 MFCCs and 12 Chroma STFT bins (librosa [6]); clips shorter than 2.5 s are zero-filled, as in the paper (“null values were replaced with zeroes”). Features are flattened and concatenated into the paper’s 3,672-dimensional vector ( $108 \times 34$ ) and standardised per dimension with training-set statistics. Each training clip contributes four versions, exactly as in the paper: the original, Gaussian noise with amplitude  $\alpha = 0.035 \cdot u \cdot \max |x|$  ( $u \sim U(0, 1)$ ), a pitch shift of 0.7 semitones, and noise followed by pitch shift. Validation and test clips are never augmented, except in the deliberately leaky protocol below.

### 3.2 Models

**Paper models.** We implement the paper’s Table 2 architecture in Keras [7]: five convolution blocks (Conv1D  $\rightarrow$  BatchNorm  $\rightarrow$  MaxPool/2) with 128, 128, 64, 64 and 32 filters, kernel sizes 5, 5, 5, 3, 3, ReLU, “same” padding, and 20% dropout after blocks 2, 4 and 5; then Flatten, Dense(128, ReLU), BatchNorm and a 7-way softmax. CNN\_Bi-LSTM inserts a bidirectional LSTM (64 units per direction, full sequence) before block 5. The **averaging ensemble** is the mean of the two models’ class probabilities, which is what the paper’s Average layer computes at inference. Our CNN\_Bi-LSTM has 688,807 parameters versus 672,423 in the paper’s Table 2; the small difference comes from the last convolution, whose parameter count in the paper (6,176) implies 64 input channels although the Bi-LSTM listed before it outputs 128.

**Training (paper Table 3).** Adam (learning rate  $10^{-3}$ ), cross-entropy, batch size 64, up to 100 epochs, early stopping on validation accuracy (patience 5, best weights restored), and learning-rate halving on a validation-accuracy plateau (patience 3, minimum  $10^{-5}$ ). All random seeds are fixed to 42.

**Baselines** (scikit-learn [8]) use the mean and standard deviation of each frame feature over time (68 values): a majority-class predictor (chance level), an RBF SVM ( $C=10$ ) and a 300-tree Random Forest. They are trained on exactly the same (augmented) training clips as the deep models.

### 3.3 Evaluation protocols

**P1 Paper protocol (H1):** stratified random 80/10/10 split of the 1,440 RAVDESS clips; augmentation applied to the 1,152 training clips only (4,608 training samples, as in the paper).

**P2 Augment-before-split (H2):** all 5,760 augmented samples are pooled and then split 80/10/10 at random, so a test clip’s augmented copies can be in the training set.

**P3 Speaker-independent (H3, H5):** 4-fold cross-validation in which each fold holds out six actors (three male, three female): actors 1–6, 7–12, 13–18, 19–24. Every clip is tested exactly once by a model that never heard that speaker; 10% of the training clips are held out for early stopping.

**P4 Cross-dataset (H4):** train on all of RAVDESS and test on all of TESS, and the reverse. No target data is used for training, early stopping or normalisation. The ablation (H6) retrain the 1D-CNN and SVM under P1 with one feature group removed, with MFCC only, and without augmentation (plus a control with longer early-stopping patience, Sec. 4.6). We report accuracy, macro-F1 (robust to RAVDESS’s larger neutral class) and, for comparison with the paper, weighted F1. The **bias check** compares accuracy on the 12 male versus the 12 female RAVDESS actors under P3 with a two-proportion  $z$ -test ( $n=720$  clips per group).

## 4 Results

Table 1 summarises all models under the four protocols; Fig. 2 visualises the three RAVDESS-based protocols and the RAVDESS $\rightarrow$ TESS transfer.

### 4.1 H1 – Replication under the paper’s protocol

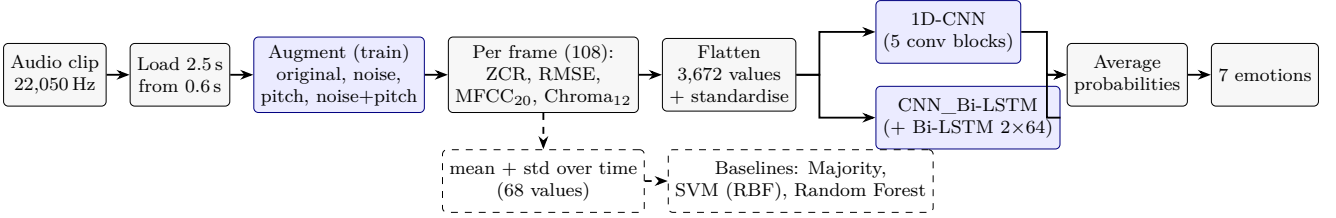
With augmentation applied to training clips only, our ensemble reaches 81.9% accuracy (weighted F1 81.9%), 15.6 points below the 97.57% reported (Table 2). The ordering of the models matches the paper (ensemble  $\geq$  CNN\_Bi-LSTM  $\geq$  1D-CNN), and all deep models beat the SVM (77.1%) and Random Forest (69.4%). **H1 is not supported:** the reported figure is not reproduced under a leak-free split.

### 4.2 H2 – Augmenting before splitting inflates accuracy

When the 5,760 augmented samples are pooled *before* the 80/10/10 split, 99.1% of the test samples have at least one augmented copy of the same recording in the training set. The ensemble then scores 96.5% (1D-CNN 95.7%), within about one percentage point of the paper’s 97.57% and 96.18%. Accuracy is 97.8% on original test clips and 95.6% on noise+pitch copies, i.e. the model recognises recordings it has effectively already seen. **H2 is supported.** The paper does not state the split order, so we cannot prove this is what happened there, but it is the only one of our settings that reproduces its numbers, and the inflation (+14.6 points for the ensemble) is much larger for the high-capacity deep models than for the SVM (+8.5), consistent with memorisation of near-duplicates [9].

### 4.3 H3 – Unseen speakers

Holding out whole actors lowers ensemble accuracy to  $58.5 \pm 1.9\%$  (macro-F1 57.8%), a drop of 23.5 points from the paper protocol. The deep models still beat the



**Figure 1:** Pipeline. Blue boxes follow the paper [1]; dashed boxes are our classical baselines, which use the same frame features summarised over time.

**Table 1:** Accuracy / macro-F1 (%) on the test sets. Speaker-independent: pooled over all 1,440 clips,  $\pm$  is the standard deviation of accuracy across the 4 speaker folds. Chance (majority class) is 20.0% on RAVDESS and 14.3% on TESS. Bold: best model per protocol, when above chance.

| Model              | Paper protocol     | Aug.-before-split  | Speaker-indep.                          | RAV→TESS           | TESS→RAV   |
|--------------------|--------------------|--------------------|-----------------------------------------|--------------------|------------|
| Majority class     | 20.1 / 4.8         | 20.0 / 4.8         | 20.0 $\pm$ 0.0 / 4.8                    | 14.3 / 3.6         | 20.0 / 4.8 |
| SVM (RBF)          | 77.1 / 76.7        | 85.6 / 85.4        | 49.1 $\pm$ 4.3 / 48.8                   | 23.9 / 17.2        | 13.6 / 7.7 |
| Random Forest      | 69.4 / 68.0        | 71.5 / 70.5        | 46.8 $\pm$ 2.6 / 43.3                   | <b>31.1 / 23.8</b> | 14.4 / 5.6 |
| 1D-CNN             | 79.2 / 78.6        | 95.7 / 95.7        | 56.7 $\pm$ 2.0 / 56.0                   | 17.8 / 14.3        | 14.1 / 6.6 |
| CNN_Bi-LSTM        | 81.2 / 80.6        | 94.4 / 94.5        | 56.5 $\pm$ 2.9 / 56.2                   | 17.9 / 17.1        | 13.6 / 4.0 |
| Averaging ensemble | <b>81.9 / 81.4</b> | <b>96.5 / 96.6</b> | <b>58.5 <math>\pm</math> 1.9 / 57.8</b> | 19.7 / 17.0        | 14.1 / 6.6 |

**Table 2:** RAVDESS replication: accuracy and weighted F1 (%).

| Model              | Reported in paper |       | Ours: paper protocol |       | Ours: aug.-before-split |       |
|--------------------|-------------------|-------|----------------------|-------|-------------------------|-------|
|                    | Acc.              | W-F1  | Acc.                 | W-F1  | Acc.                    | W-F1  |
| 1D-CNN             | 96.18             | 96.22 | 79.17                | 79.04 | 95.66                   | 95.65 |
| CNN_Bi-LSTM        | 97.57             | 97.29 | 81.25                | 81.14 | 94.44                   | 94.44 |
| Averaging ensemble | 97.57             | 97.56 | 81.94                | 81.94 | 96.53                   | 96.53 |

best classical baseline (SVM, 49.1%) by 9.4 points, so they do learn transferable cues, but performance varies strongly between speakers: from 40.0% (Actor 09) to 81.7% (Actor 08). Recall is highest for angry (73.4%) and lowest for sad (34.4%); the most frequent errors are sad→neutral (27.6% of sad clips), neutral→sad (13.5%) and happy→fearful (13.5%) (Fig. 3). **H3 is supported.**

#### 4.4 H4 – Unseen dataset

Cross-corpus transfer fails in both directions. Trained on RAVDESS, the ensemble reaches only 19.7% on TESS (chance 14.3%) and labels 53.1% of all TESS clips as *sad*. Trained on TESS, it scores 14.1% on RAVDESS, *below* the 20.0% majority baseline, and predicts *fearful* for 87.9% of clips. Per-class recall on TESS (RAVDESS-trained ensemble) is:

| Ang. | Dis. | Fea. | Hap. | Neu. | Sad. | Sur. |
|------|------|------|------|------|------|------|
| 37.0 | 29.2 | 20.2 | 0.8  | 0.2  | 48.5 | 2.0  |

Interestingly, the Random Forest transfers best (31.1%). Its time-pooled statistics do not depend on *where* in the 2.5 s window a sound occurs, whereas the paper’s flattened vector ties every value to a fixed frame position. **H4 is supported.**

**Controlled check: zero padding.** TESS clips are short, so after the 0.6 s offset 42% of TESS frames are zero padding, against 0% for RAVDESS. Silent frames

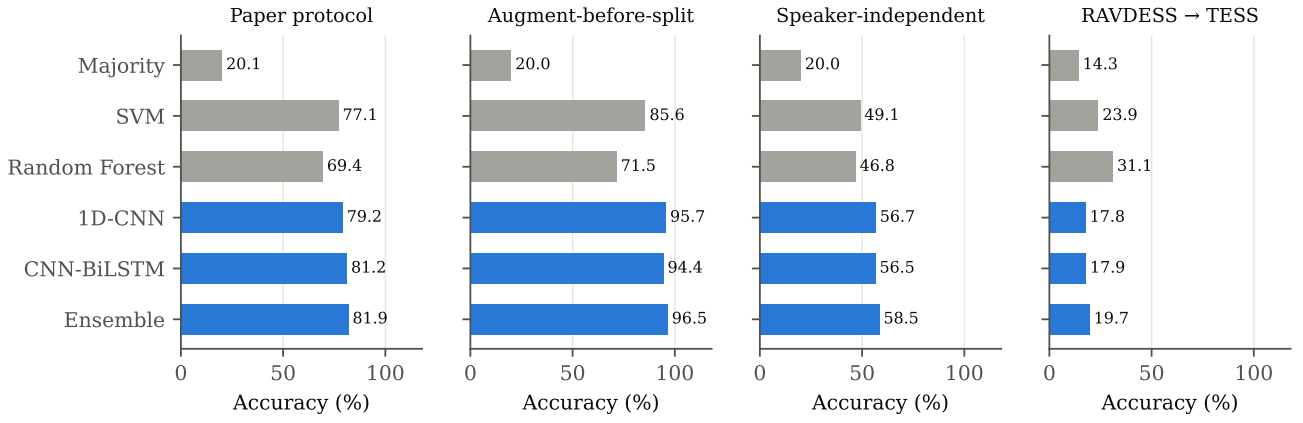
**Table 3:** Bias check, speaker-independent protocol (720 clips per gender). Gap = female – male.

| Model              | Acc. male | Acc. female | Gap (pp) | $z$   | $p$    |
|--------------------|-----------|-------------|----------|-------|--------|
| SVM (RBF)          | 45.3      | 52.9        | +7.6     | -2.90 | 0.004  |
| Random Forest      | 41.0      | 52.6        | +11.7    | -4.44 | <0.001 |
| 1D-CNN             | 49.4      | 63.9        | +14.4    | -5.53 | <0.001 |
| CNN_Bi-LSTM        | 47.2      | 65.8        | +18.6    | -7.12 | <0.001 |
| Averaging ensemble | 50.7      | 66.2        | +15.6    | -5.99 | <0.001 |

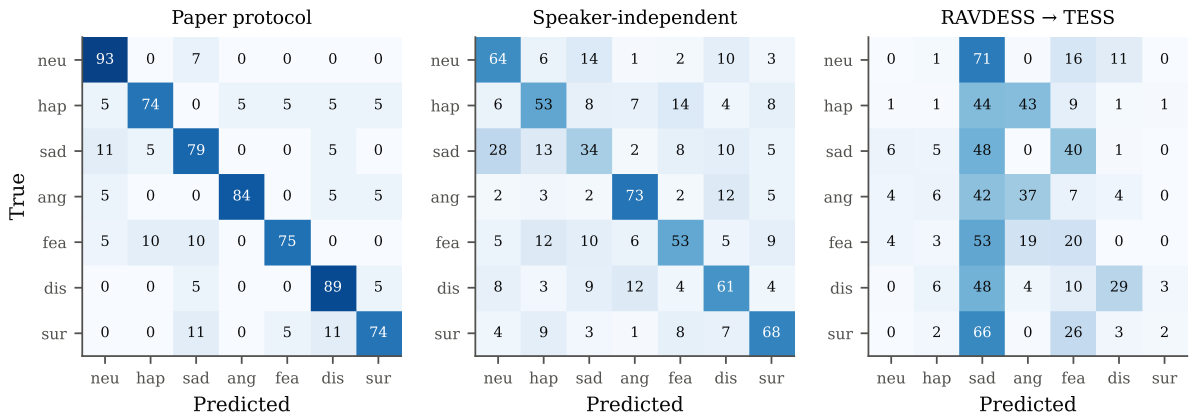
have near-zero energy and look like low-arousal speech. To test this without retraining, we re-extracted TESS with offset 0 s (18% padding) and evaluated the *same* RAVDESS-trained models: accuracy rises from 19.7% to 30.5% and the share of “sad” predictions falls from 53% to 11%. Padding therefore explains part of the collapse, but even without it the model remains far below its within-corpus accuracy, pointing to the other differences between corpora: speakers, single words vs. sentences, and recording conditions. The older TESS speaker is recognised worse (15.5% vs. 23.9% for the younger one).

#### 4.5 H5 – Gender bias check

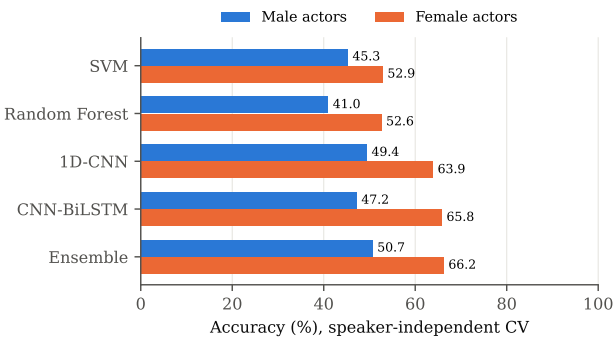
Every model is more accurate for female than for male actors (Table 3, Fig. 4). For the ensemble the gap is +15.6 points (66.2% vs. 50.7%,  $p < 0.001$  by a two-proportion  $z$ -test). Because clips of the same actor are not independent, we also compared the 24 per-actor accuracies: the female actors are still clearly better (Mann–Whitney  $U = 129$  of a maximum 144,  $p = 0.001$ ). The data are balanced (12 actors and 720 clips per gender), so this is not a sampling artefact of RAVDESS’s composition. The gap is largest for surprised, neutral and disgust and is the only one reversed for happy (Table 4). **H5 is supported:** a model deployed from this pipeline would systematically serve male speakers worse.



**Figure 2:** Accuracy of every model as the protocol becomes stricter (left to right). Grey: classical baselines; blue: the paper’s models. The same ensemble ranges from 96.5% to 19.7%.



**Figure 3:** Row-normalised confusion matrices (%) of the ensemble. Under the paper protocol most mass is on the diagonal; for unseen speakers low-arousal emotions (neutral, sad) are confused; on TESS predictions collapse onto a few classes.



**Figure 4:** Accuracy by speaker gender (speaker-independent CV).

#### 4.6 H6 – Feature and augmentation ablation

Table 5 retrains the 1D-CNN and the SVM on the paper-protocol split with parts of the input removed. The test set has 144 clips, so one clip is 0.69 points and the standard error of an accuracy near 79% is about 3.4 points; differences smaller than roughly twice that are not meaningful.

- **MFCC is essential.** Removing the 20 MFCCs changes accuracy by  $-22.2$  points for the 1D-CNN

**Table 4:** Per-emotion recall (%) of the ensemble by gender.

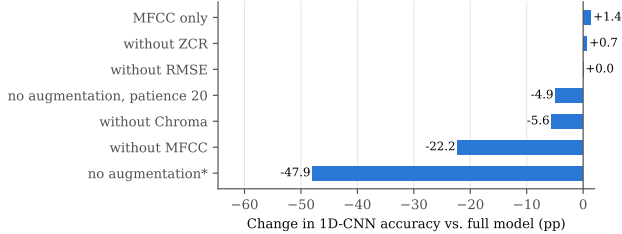
| Emotion   | Recall male | Recall female | Gap (pp) |
|-----------|-------------|---------------|----------|
| Neutral   | 54.2        | 74.3          | +20.1    |
| Happy     | 54.2        | 52.1          | −2.1     |
| Sad       | 30.2        | 38.5          | +8.3     |
| Angry     | 67.7        | 79.2          | +11.5    |
| Fearful   | 43.8        | 61.5          | +17.7    |
| Disgust   | 51.0        | 70.8          | +19.8    |
| Surprised | 52.1        | 83.3          | +31.2    |

and  $-31.9$  for the SVM, the only feature removal clearly beyond noise. *MFCC alone* (80.6%) is at least as accurate as the full feature set.

- **ZCR and RMSE add nothing measurable** ( $\Delta$  of  $+0.7$  and  $+0.0$  points).
- **Chroma: inconsistent.** Removing Chroma changes the 1D-CNN by  $-5.6$  points but the SVM by  $+2.1$  (opposite directions). Both changes are within about two standard errors, so we find no robust evidence for the paper’s LIME-based claim that Chroma contributes substantially.
- **Augmentation helps modestly, and interacts with early stopping.** Without augmentation, training stops at epoch 9 with validation accuracy still

**Table 5:** Ablation on the paper-protocol split (accuracy %, 144 test clips).  $\Delta$  is relative to the full 1D-CNN (79.2%). \*Training stopped at epoch 9 with validation accuracy at chance level (see text).

| Variant                                | Input dim. | SVM acc. | 1D-CNN acc. | $\Delta$ CNN (pp) |
|----------------------------------------|------------|----------|-------------|-------------------|
| All features (ZCR+RMSE+MFCC+Chroma)    | 3,672      | 77.1     | 79.2        | —                 |
| without ZCR                            | 3,564      | 77.1     | 79.9        | +0.7              |
| without RMSE                           | 3,564      | 77.1     | 79.2        | +0.0              |
| without MFCC                           | 1,512      | 45.1     | 56.9        | -22.2             |
| without Chroma                         | 2,376      | 79.2     | 73.6        | -5.6              |
| MFCC only                              | 2,160      | 78.5     | 80.6        | +1.4              |
| No augmentation (paper early stopping) | 3,672      | 78.5     | 31.2        | -47.9             |
| No augmentation, patience 20           | 3,672      | —        | 74.3        | -4.9              |



**Figure 5:** Change in 1D-CNN accuracy when parts of the input are removed (paper-protocol split). \*No augmentation with the paper’s early stopping, which halts before the model learns; see text.

at chance, giving 31.2%. The cause is that an un-augmented epoch has  $4\times$  fewer updates (18 vs. 72 batches), so the paper’s patience of 5 epochs is too short. With a patience of 20 epochs, the un-augmented model reaches 74.3%, so dropping augmentation changes accuracy by  $-4.9$  points (about 1.4 standard errors), not  $-47.9$ . Augmentation therefore helps modestly at best on this split.

**H6 is partly supported:** MFCC carries almost all the information; a positive contribution of Chroma is not confirmed. A practical corollary, on this split, is that a 2,160-value MFCC-only input gives a smaller model with no loss in accuracy.

## 4.7 Efficiency

The 1D-CNN has 616,615 parameters and the CNN\_Bi-LSTM 688,807 (ensemble 1,305,422). On a 4-core CPU, training under the paper protocol took 10.6 and 13.1 minutes (21 and 23 epochs before early stopping), and inference costs 2.70 and 4.14 ms per clip. The SVM trains in seconds and predicts in 0.26 ms per clip, and it is only 4.9 points behind the ensemble under the paper protocol, which puts the value of the deep ensemble in perspective.

## 5 Error Analysis and Discussion

**Why the paper’s protocol overestimates performance.** Emotion corpora such as RAVDESS contain a small number of speakers who each record the same two sentences many times. A random split therefore places recordings of the *same actor saying the same sentence* in both training and test sets, and augmentation before the split adds near-copies of the *same recording*. Both violate the assumption that test data are independent of

training data. The 97% figure measures recognition of familiar voices (or familiar recordings), not of emotion.

**Why the models fail on new speakers.** The input is a globally standardised vector of absolute feature values. MFCC and Chroma levels depend strongly on vocal-tract shape and fundamental frequency, i.e. on *who* is speaking; nothing in the pipeline removes speaker identity, so a model can reach high within-speaker accuracy by learning speaker-specific baselines. For a new speaker those baselines are wrong. The dominant errors (sad $\rightarrow$ neutral, neutral $\rightarrow$ sad) are between low-arousal classes whose differences are subtle relative to between-speaker variation. Merging RAVDESS *calm* into *neutral*, as the paper does, makes the neutral class more heterogeneous and adds to this confusion.

**Why women are recognised better.** We can only offer hypotheses: female voices typically have a higher and wider pitch range, which Chroma and MFCC capture, so emotional pitch changes may be more separable; and the augmentation (pitch shift of +0.7 semitones) never moves a voice across the male–female range. Speaker-level normalisation or gender-balanced model selection would be the first mitigations to test.

**Why cross-corpus transfer collapses.** Besides speakers, the corpora differ in utterance type (a full sentence vs. “Say the word \_\_\_”), duration (hence padding), speaker age and recording setup. The flattened, position-dependent representation amplifies these differences: the same emotion expressed 0.3 s later produces a different input vector. The controlled padding check shows that one of these factors alone moves accuracy by 10.8 points.

**Take-away.** The paper’s architecture is a reasonable lightweight model: it beats classical baselines under every within-corpus protocol. But its reported accuracy is a property of the evaluation protocol as much as of the model. For deployment, the speaker-independent figure (58.5%) and the cross-corpus figure (near chance) are the relevant ones.

## 6 Challenges Faced

- **Under-specified details.** The paper does not state whether augmentation precedes the split, how the ensemble is trained, or which label is used for RAVDESS *calm*. We inferred the latter from its Table 1, trained the members separately and averaged them, and made the split order an explicit experiment (P2).
- **Inconsistent parameter table.** The paper’s Table 2 lists 6,176 parameters for the last convolution, which does not match a 128-channel Bi-LSTM output; we followed the layer description.
- **Compute.** Convolution of a 3,672-step input with 128 filters is expensive on a CPU: one epoch takes about 30 s for the 1D-CNN and longer for the Bi-LSTM variant, so the 23 training runs of this study took roughly 4 hours. We therefore ran the ablation on one split rather than four folds.

- **Duplicated files.** Copies of both datasets can contain duplicated folders; without de-duplication identical files would appear in both training and test sets.
- **Different clip lengths.** TESS clips are about 2s long, so after the paper’s 0.6s offset 42% of TESS frames are zero padding versus none for RAVDESS, an input shift that matters for cross-dataset transfer (Sec. 5).

## 7 Limitations and Future Work

Both corpora contain acted, scripted North-American English speech, so even our strictest protocols overestimate performance on spontaneous speech. TESS has only two female speakers, so the reverse cross-dataset direction trains on two voices, and the gender analysis is restricted to RAVDESS. We used one random seed per configuration; fold-level standard deviations indicate variance across speakers but not across initialisations. Natural next steps are speaker-level feature normalisation, domain-adversarial training, multi-corpus training, and adding CREMA-D (91 speakers) to test transfer to a larger, more diverse corpus.

## 8 Conclusion

We re-implemented a recent lightweight SER ensemble and asked how far its near-perfect accuracy generalises. Five findings stand out. (1) Under a leak-free random split our implementation reaches 81.9%, not 97.57%. (2) Applying augmentation before splitting reproduces the reported level (96.5%) because almost every test sample has a near-copy in training. (3) For unseen speakers accuracy drops to 58.5%, still well above classical baselines. (4) Across corpora the model is near chance, partly because the paper’s fixed offset leaves 42% of TESS frames as zero padding. (5) The model is significantly less accurate for male speakers (+15.6 points). For practitioners the lesson is concrete: report speaker-independent and cross-corpus results, split before augmenting, and check performance per demographic group before deploying an emotion recogniser.

**Reproducibility.** All code, the exact library versions and the scripts that generate every number and figure in this report are in the project repository (`README.md`); the pipeline runs end-to-end with four commands.

## References

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